

DSA02-CO2-AT2

Code of Conduct:

I G. Gopi Chand (192424384) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary

action.

Signature of the student:

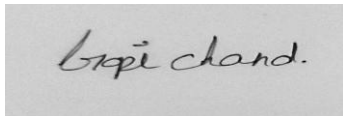
A rectangular box containing a handwritten signature in cursive script that reads "Gopi Chand."

Image Processing – Scenario-Based Questions (Simple & Clear Answers)

1. Low Contrast Image

(a) Interpret the Scenario

A grayscale image looks dull because the object and background have nearly the same intensity values. As a result, it is difficult to distinguish objects from the background.

Cause:

- Pixel values are concentrated in a small intensity range.
 - There is little difference between dark and bright regions.
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(b) Key Issues Affecting Image Visibility

- Low image contrast

- Poor visibility of objects
 - Loss of important details
 - Difficult image analysis and segmentation
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(c) Appropriate Enhancement Technique

Technique: Histogram Equalization

Working:

- Redistributes pixel intensity values over the full intensity range.
 - Increases contrast by spreading the histogram.
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(d) Justification

Histogram Equalization is suitable because:

- Improves overall image contrast.
 - Reveals hidden details.
 - Makes objects easier to identify.
 - Simple and widely used enhancement technique.
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(e) Structured Answer

Item	Explanation
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Problem	Low contrast between object and background
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Cause	Narrow intensity range
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Issues	Poor visibility and detail loss
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Technique	Histogram Equalization
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Result	Better contrast and clearer image
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2. Uneven Illumination

(a) Interpret the Scenario

The image contains bright and dark regions because lighting is not uniform across the scene.

Impact:

- Some areas appear too bright.
 - Some areas appear too dark.
 - Image analysis becomes difficult.
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(b) Key Factors Causing Intensity Variation

- Non-uniform lighting
- Shadows
- Uneven camera exposure
- Different light source positions

(c) Suitable Enhancement Method

Technique: Homomorphic Filtering

Working:

- Separates illumination and reflectance components.
 - Reduces lighting variations.
 - Enhances image contrast.
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(d) Justification

Homomorphic filtering is preferred because:

- Corrects uneven illumination.
 - Improves image brightness uniformly.
 - Preserves object details.
 - Produces natural-looking images.
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(e) Structured Answer

Item	Explanation
Problem	Uneven lighting
Cause	Non-uniform illumination
Issues	Bright and dark patches
Technique	Homomorphic Filtering
Result	Uniform brightness and improved visibility

3. Salt-and-Pepper Noise

(a) Explain the Scenario

A scanned document contains random black and white pixels, making the text difficult to read.

Characteristics

- Random white pixels (Salt)
 - Random black pixels (Pepper)
 - Appears as isolated dots
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(b) Key Issues

- Reduced readability
 - Image distortion
 - Poor OCR (Optical Character Recognition)
 - Loss of image quality
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(c) Appropriate Filtering Technique

Technique: Median Filter

Working

- Replaces each pixel with the median of neighboring pixels.
 - Removes isolated noisy pixels without blurring edges.
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(d) Justification

Median Filter is better because:

- Removes salt-and-pepper noise effectively.
 - Preserves edges.
 - Better than averaging filter, which blurs the image.
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(e) Structured Answer

Item	Explanation
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Problem	Salt-and-pepper noise
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Cause	Random black and white pixels
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Item	Explanation
Issues	Poor readability
Technique	Median Filter
Result	Clean image with preserved edges

4. Blurred Edges

(a) Interpret the Scenario

The image was heavily smoothed to remove noise, but this also removed important edge information.

Reason

- Smoothing averages neighboring pixels.
 - Sharp intensity changes become blurred.
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(b) Key Issues

- Loss of edge information
 - Reduced sharpness
 - Poor object boundaries
 - Difficult object detection
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(c) Appropriate Technique

Technique: Laplacian Sharpening (*or Unsharp Masking*)

Working

- Enhances intensity differences.
 - Highlights object boundaries.
 - Restores sharp edges.
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(d) Justification

Sharpening is suitable because:

- Restores lost edge information.
 - Improves image clarity.
 - Makes object boundaries more visible.
 - Maintains important image details.
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(e) Structured Answer

Item	Explanation
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Problem	Blurred edges
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Cause	Excessive smoothing
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Issues	Edge loss and poor sharpness
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Technique	Laplacian Sharpening
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Result	Clear and sharp edges
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5. High-Frequency Noise

(a) Interpret the Scenario

A satellite image contains fine-grained random noise that affects the visibility of roads, buildings, and other features.

High-frequency noise

- Rapid intensity changes.
 - Small random pixel variations.
 - Appears as grainy texture.
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(b) Key Issues

- Poor image clarity
- Difficult feature extraction
- Reduced object visibility

- Lower analysis accuracy
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(c) Appropriate Frequency Domain Filtering Method

Technique: Low-Pass Filter (LPF)

Working

- Removes high-frequency noise.
 - Preserves low-frequency image information.
 - Produces a smoother image.
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(d) Justification

Low-Pass Filter is preferred because:

- Removes high-frequency noise effectively.
 - Improves image quality.
 - Makes important features easier to identify.
 - Commonly used in satellite image processing.
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(e) Structured Answer

Item	Explanation
Problem	High-frequency noise
Cause	Random fine-grained intensity changes
Issues	Poor clarity and feature visibility
Technique	Low-Pass Filter
Result	Noise reduction and improved image quality

Quick Revision Table

Question	Problem	Best Technique	Main Benefit
1	Low Contrast	Histogram Equalization	Improves contrast
2	Uneven Illumination	Homomorphic Filtering	Normalizes brightness
3	Salt-and-Pepper Noise	Median Filter	Removes impulse noise while preserving edges
4	Blurred Edges	Laplacian Sharpening (or Unsharp Masking)	Restores sharp edges
5	High-Frequency Noise	Low-Pass Filter	Reduces high-frequency noise